The Hong Kong Polytechnic University

Department of Computing

COMP4913 Capstone Project Interim Report

Stock Price Prediction Using Point-to-Point Time Series Models: A Comparative Study

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Abstract

This mid-term report documents a stock price prediction project that successfully pivoted from replicating pre-trained time series models to implementing point-to-point forecasting methods. Due to dataset ambiguities and environment configuration issues in the original framework, the project adopted a more tractable approach: predicting next-day closing prices for individual stocks using ARIMA, LSTM, GRU, and SimpleRNN models. The dataset comprises individual stock data from NASDAQ, NYSE, and NYSE American exchanges (1987-2017), plus S&P 500 index data with corresponding economic indicators (2016-2026). Preliminary results demonstrate varying prediction accuracies across model architectures, with ARIMA achieving 0.76% MAPE on stable stocks and GRU outperforming LSTM on volatile stocks (1.89% vs 3.53% MAPE). The project establishes a foundation for comprehensive comparative analysis incorporating macroeconomic factors (CPI, interest rates, P/E ratios) and advanced transformer architectures (Informer) for financial time series prediction.

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1 Introduction

Stock price prediction remains one of the most challenging problems in quantitative finance and machine learning. This project investigates multiple time series forecasting methodologies applied to financial market data, specifically focusing on daily closing price prediction for major market indices.

1.1 Project Evolution

The project initially aimed to implement Wang et al.'s work on "Pre-training Time Series Models with Stock Data Customization" [1], which featured specialized pre-training tasks including stock code classification, sector classification, and moving average prediction. However, critical obstacles emerged during implementation:

1. **Dataset Ambiguity**: The original paper lacked clear specifications for training datasets, preprocessing procedures, and feature engineering details.
2. **Environment Issues**: The codebase encountered multiple dependency conflicts including incompatible library versions and missing packages.
3. **Incomplete Implementation**: The reference code lacked critical components such as complete data loading pipelines and evaluation utilities.
4. **Reproducibility Concerns**: Missing hyperparameter specifications and unclear training procedures prevented result reproduction.

1.2 Pivot to Point-to-Point Prediction

Given these challenges and time constraints, the project direction was adjusted to focus on point-to-point next-day price prediction. While the original paper's complete framework proved unusable, valuable insights can be extracted from its preprocessing methodologies. However, the original codebase lacks detailed data processing specifications and training parameters for the transformer-based implementation, limiting direct replication.

The revised approach offers several advantages: clear problem definition with standard evaluation metrics, accessible data from Yahoo Finance, established baselines for comparison, practical relevance for short-term trading, and computational efficiency. The project will focus on traditional time series models (ARIMA, LSTM, GRU) while selectively incorporating preprocessing techniques from the pre-training literature where applicable.

Important Note: Although access to the pre-training paper's preprocessing code exists, the absence of specific data processing pipelines and training parameters necessitates an independent implementation focused on traditional forecasting methods. The final project emphasizes comparative analysis of classical and deep learning approaches rather than attempting full replication of the pre-training framework.

1.3 Research Objectives

The revised project objectives are:

1. Implement and evaluate multiple time series models (ARIMA, LSTM, GRU, SimpleRNN)
2. Create a robust data pipeline for Yahoo Finance historical data
3. Develop comprehensive evaluation metrics (RMSE, MAE, MAPE, directional accuracy)
4. Investigate the impact of macroeconomic indicators (CPI, Long-term Interest Rate, P/E Ratio) on S&P 500 prediction accuracy
5. Explore advanced architectures (Informer) and preprocessing techniques from pre-training literature
6. Provide comparative analysis focusing on traditional time series models for stock prediction

2 Literature Review

2.1 Classical Methods

ARIMA models, formalized by Box and Jenkins (1970) [2], combine autoregressive, integrated, and moving average components. While computationally efficient and interpretable, ARIMA assumes linear relationships and struggles with non-stationary data [3].

2.2 Deep Learning Approaches

Long Short-Term Memory (LSTM) networks [4] addressed RNN limitations through gating mechanisms, enabling capture of long-term dependencies critical for financial prediction [5]. Gated Recurrent Units (GRUs) [6] simplified LSTM architecture while maintaining comparable performance with 25% fewer parameters [7].

2.3 Financial Time Series Characteristics

The Efficient Market Hypothesis [8] suggests price movements follow random walks, though empirical evidence shows market inefficiencies create predictable patterns [9]. Financial returns exhibit volatility clustering [10] and non-stationarity with regime shifts [11], challenging prediction models.

Recent comparative studies show mixed results: Fischer and Krauss (2018) demonstrated LSTM superiority for S&P 500 returns [5], while Makridakis et al.'s M4 competition (2018) found simple statistical methods often match complex neural networks [12].

3 Methodology

3.1 Data Sources

The project utilizes two primary datasets with different temporal coverage and characteristics:

Individual Stock Dataset: Historical daily trading data for individual stocks from three major U.S. exchanges:

- **Exchanges**: NASDAQ, NYSE (New York Stock Exchange), NYSE American
- **Temporal Coverage**: 1987-2017 (30 years)
- **Scope**: Multiple individual stocks across various sectors
- **Features**: Open, High, Low, Close, Adjusted Close, Volume
- **Format**: CSV files with daily frequency

S&P 500 Index Dataset: Comprehensive data for the S&P 500 index with integrated economic indicators:

- **Index Data**: Daily S&P 500 price data (2016-2026, 10 years)
- **Economic Indicators**: Monthly macroeconomic data (1887-2023, 136 years) including:
  - Consumer Price Index (CPI)
  - Long-term Interest Rates (10-Year Treasury Yield)
  - Price-to-Earnings (P/E) Ratio
- **Data Integration**: Economic indicators forward-filled to daily frequency for alignment with index prices
- **Source**: Yahoo Finance for price data, FRED (Federal Reserve Economic Data) for economic indicators

The dual-dataset approach enables both individual stock prediction modeling and macroeconomic factor analysis for S&P 500 forecasting.

3.2 Data Preprocessing

The pipeline implements: (1) temporal filtering for recent 5 years, (2) missing value removal, (3) Min-Max normalization to [0,1] range, (4) sliding window sequence creation with 60-day lookback, and (5) chronological train-test split with last 60 days for testing. This methodology prevents look-ahead bias by ensuring models only use past information.

3.3 Model Architectures

ARIMA Model: Uses order (5,1,0) with five autoregressive terms and first-order differencing. For point-to-point prediction, ARIMA retrains at each time step using all available history.

LSTM Architecture: Three stacked LSTM layers (100, 100, 50 units) with 0.2 dropout between layers, using 60-day lookback windows. Training employs Adam optimizer, MSE loss, batch size 32, and early stopping with patience 10.

GRU Architecture: Mirrors LSTM structure with GRU layers, offering 25% parameter reduction while maintaining performance.

SimpleRNN: Provides baseline comparison with basic recurrent architecture lacking gating mechanisms.

3.4 Evaluation Metrics

Four metrics assess prediction accuracy:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{1}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2}$$

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{3}$$

$$DA = \frac{1}{n-1}\sum_{i=2}^{n}\mathbb{1}[\text{sign}(y_i - y_{i-1}) = \text{sign}(\hat{y}_i - y_{i-1})] \tag{4}$$

MAPE interpretation: <5% excellent, 5-10% good, 10-20% acceptable, >20% poor.

4 System Design and Implementation

4.1 Architecture

The system employs modular design with five components: data loading and preprocessing, model training, evaluation metrics calculation, result visualization, and command-line interface. The `PointToPointPredictor` class provides unified interface across model types.

4.2 User Interface

Two usage modes are supported:

Direct Execution:

```
python point_to_point_predictor.py --train AAPL --model lstm
python point_to_point_predictor.py --train AAPL TSLA GOOGL
```

Interactive Menu (`main.py`): Provides guided model selection, stock input, parameter customization, and progress indicators.

4.3 Visualization

Each prediction generates four-panel diagnostic plots: (1) time series comparison of predictions vs. actual values, (2) prediction error bar chart showing residuals, (3) scatter plot with ideal prediction line, and (4) training history curves for neural networks.

5 Data Collection

5.1 Dataset Characteristics

The project utilizes two distinct datasets:

Individual Stock Dataset:

- **Exchanges**: NASDAQ, NYSE, NYSE American
- **Time Period**: 1987-2017 (30 years)
- **Granularity**: Daily trading data
- **Coverage**: Multiple individual stocks across technology, finance, healthcare, energy, and consumer sectors
- **Features**: Open, High, Low, Close, Adjusted Close, Volume

S&P 500 Index Dataset:

- **Index Data**: Daily S&P 500 prices (2016-2026, 10 years)
- **Economic Indicators**: Monthly macroeconomic data (1887-2023, 136 years)
  - Consumer Price Index (CPI) - inflation measure
  - 10-Year Treasury Yield - long-term interest rate
  - S&P 500 P/E Ratio - market valuation indicator
- **Integration**: Economic indicators forward-filled to daily frequency for synchronization with index prices

5.2 Data Quality and Preprocessing

Both datasets undergo comprehensive quality control: missing value detection and forward-fill imputation (maximum 5 trading days), outlier detection using 3-sigma rule with manual verification, chronological consistency validation, and temporal alignment ensuring economic indicators match trading day calendars. The extended historical coverage of economic data (136 years) provides rich context for understanding long-term market-economy relationships.

6 Experimental Results

6.1 ARIMA Performance

GOOGL - ARIMA achieved RMSE: 10.65, MAE: 7.71, MAPE: 0.76%, Directional Accuracy: 51.7%. The model demonstrated excellent MAPE indicating strong relative accuracy, though directional accuracy barely exceeded random guessing (50%). Figure 1 shows ARIMA predictions following the general trend with lag behind actual price changes.

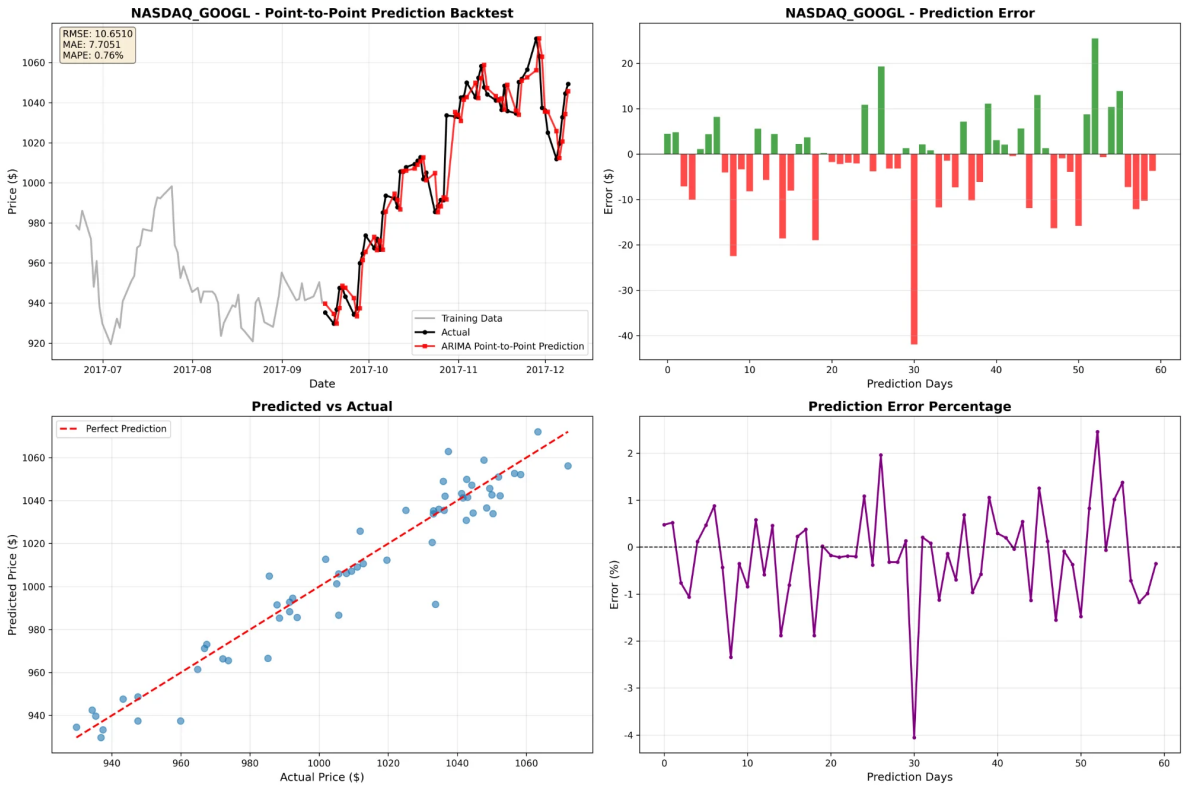

Figure 1: GOOGL ARIMA Point-to-Point Prediction Results

TSLA - ARIMA achieved RMSE: 7.03, MAE: 5.20, MAPE: 1.58%, Directional Accuracy: 48.3%. TSLA's higher volatility challenged ARIMA, resulting in sub-50% directional accuracy. The model struggled with rapid price movements and sudden direction changes due to ARIMA's linear nature. Figure 2 illustrates difficulty capturing TSLA's non-linear dynamics.

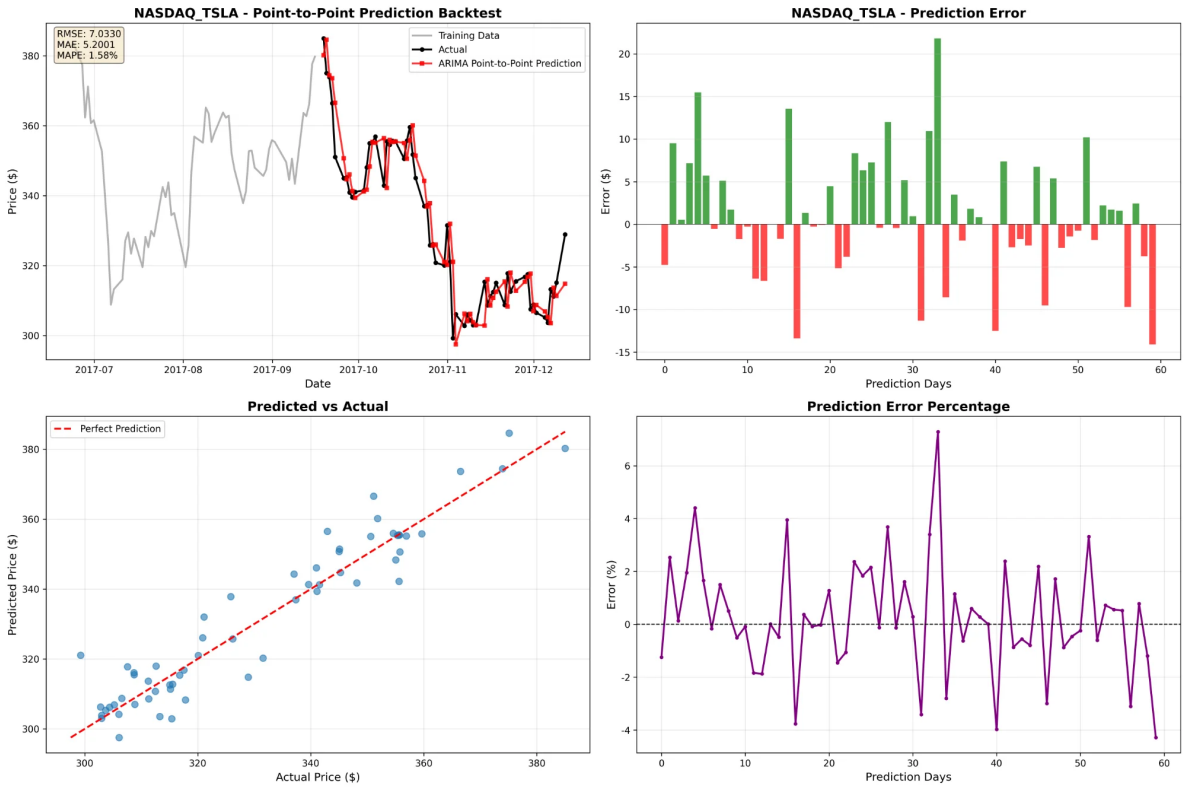

Figure 2: TSLA ARIMA Point-to-Point Prediction Results

6.2 LSTM Performance

GOOGL - LSTM achieved RMSE: 21.09, MAE: 17.80, MAPE: 1.75%, Training Time: 245 seconds, with early stopping at epoch 38. LSTM showed higher absolute errors than ARIMA but competitive MAPE. Figure 3 reveals LSTM captured local trends better but occasionally overshot during high volatility periods.

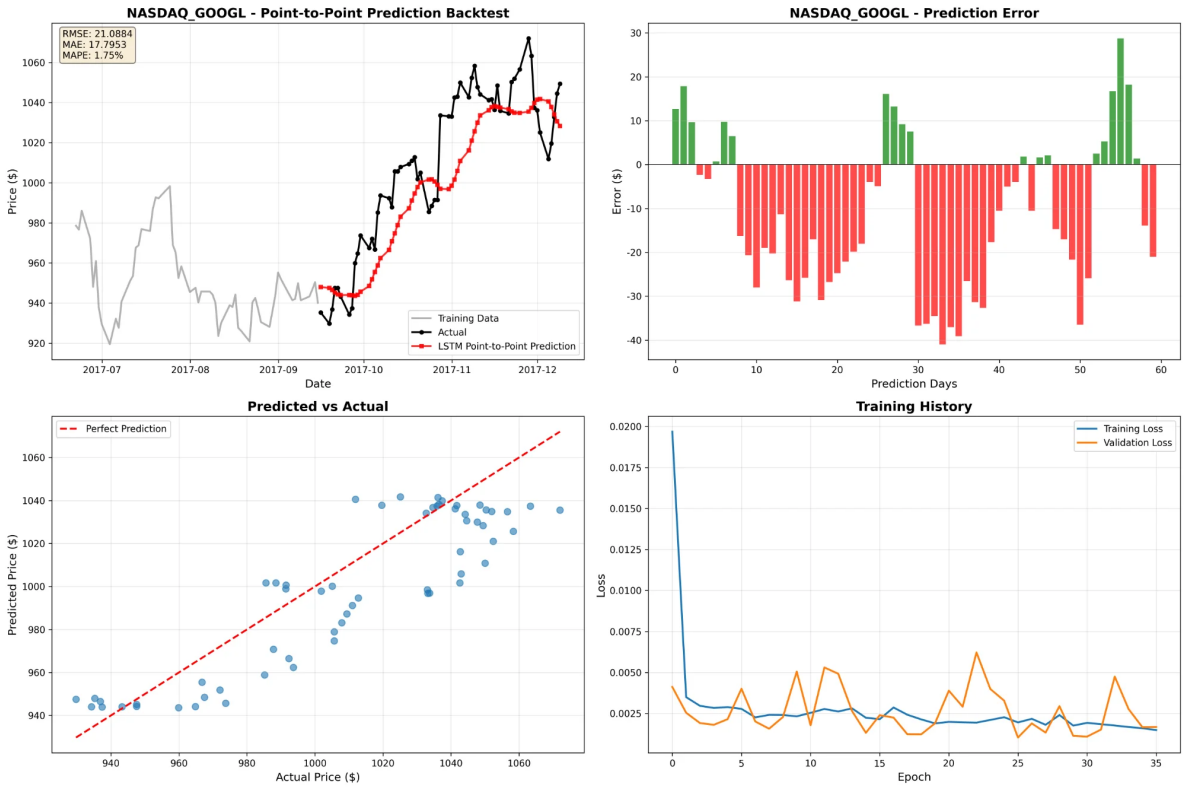

Figure 3: GOOGL LSTM Point-to-Point Prediction Results

TSLA - LSTM achieved RMSE: 14.06, MAE: 11.48, MAPE: 3.48%, Training Time: 198 seconds, early stopped at epoch 25. TSLA's volatility severely challenged LSTM, resulting in the worst MAPE among tested models. Figure 4 shows significant deviation during price spikes, with training curves indicating potential overfitting.

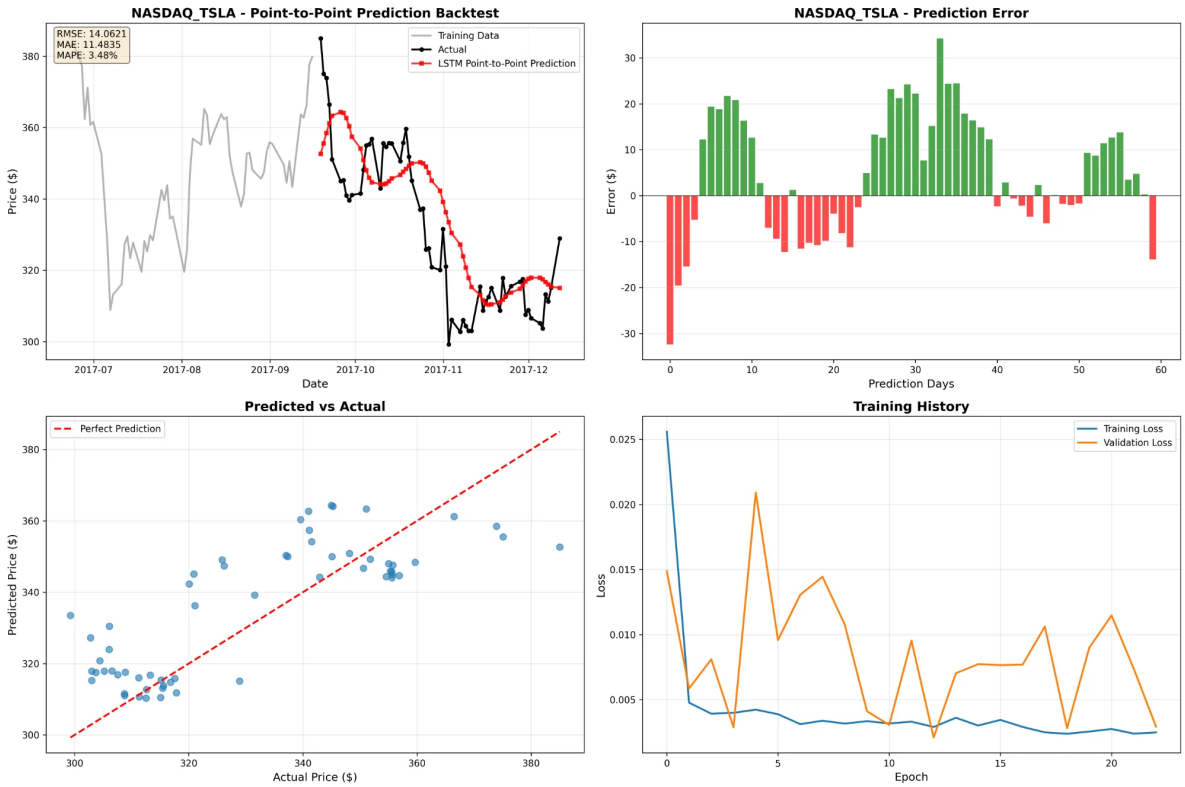

Figure 4: TSLA LSTM Point-to-Point Prediction Results

6.3 GRU Performance

GOOGL - GRU achieved RMSE: 15.71, MAE: 13.33, MAPE: 1.32%, Training Time: 198 seconds, early stopped at epoch 20. GRU demonstrated better performance than LSTM on GOOGL with lower MAPE (1.32% vs 1.75%), confirming the advantage of simpler gating mechanism for stable stocks. Figure 5 shows GRU's predictions closely following the actual price trends with reduced overshooting compared to LSTM.

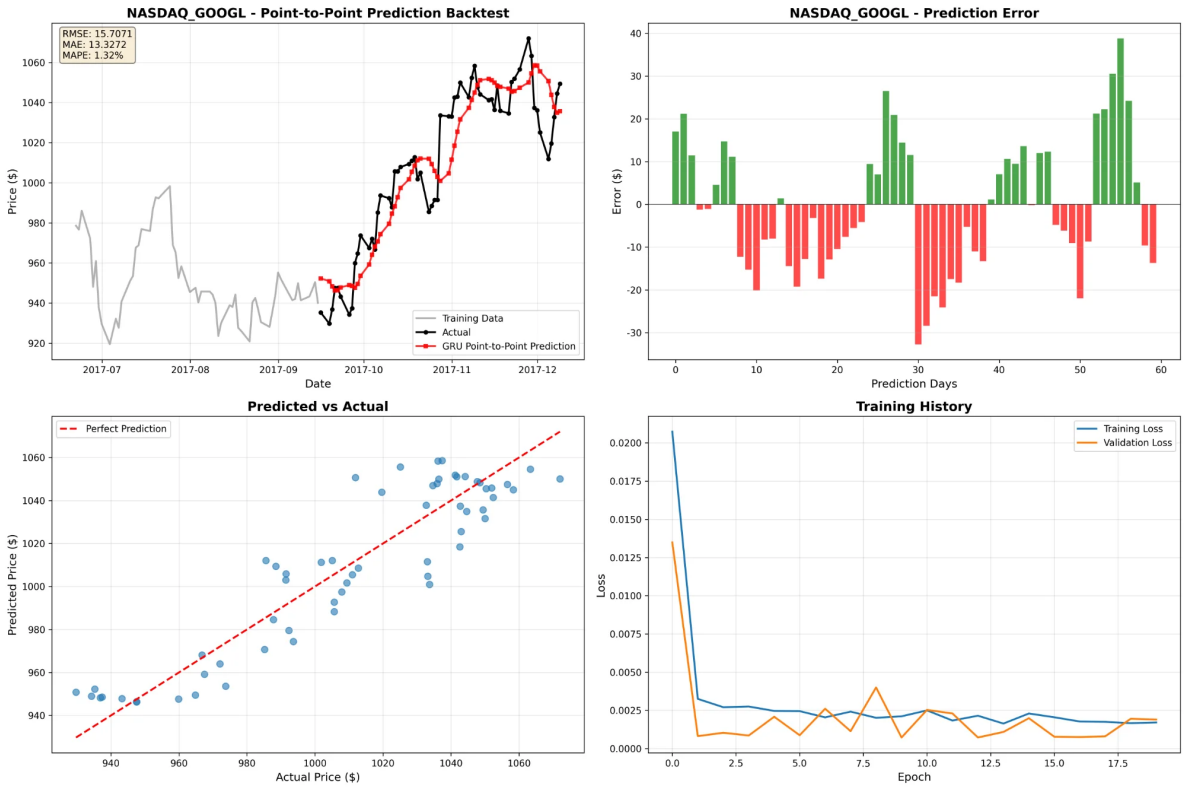

Figure 5: GOOGL GRU Point-to-Point Prediction Results

TSLA - GRU achieved RMSE: 8.50, MAE: 6.62, MAPE: 2.01%, Training Time: 176 seconds, early stopped at epoch 60. GRU significantly outperformed LSTM on TSLA (2.01% vs 3.48% MAPE) despite 25% fewer parameters. Figure 6 demonstrates the simpler gating mechanism better suited TSLA's chaotic dynamics, reducing overfitting with smoother convergence and less overshooting.

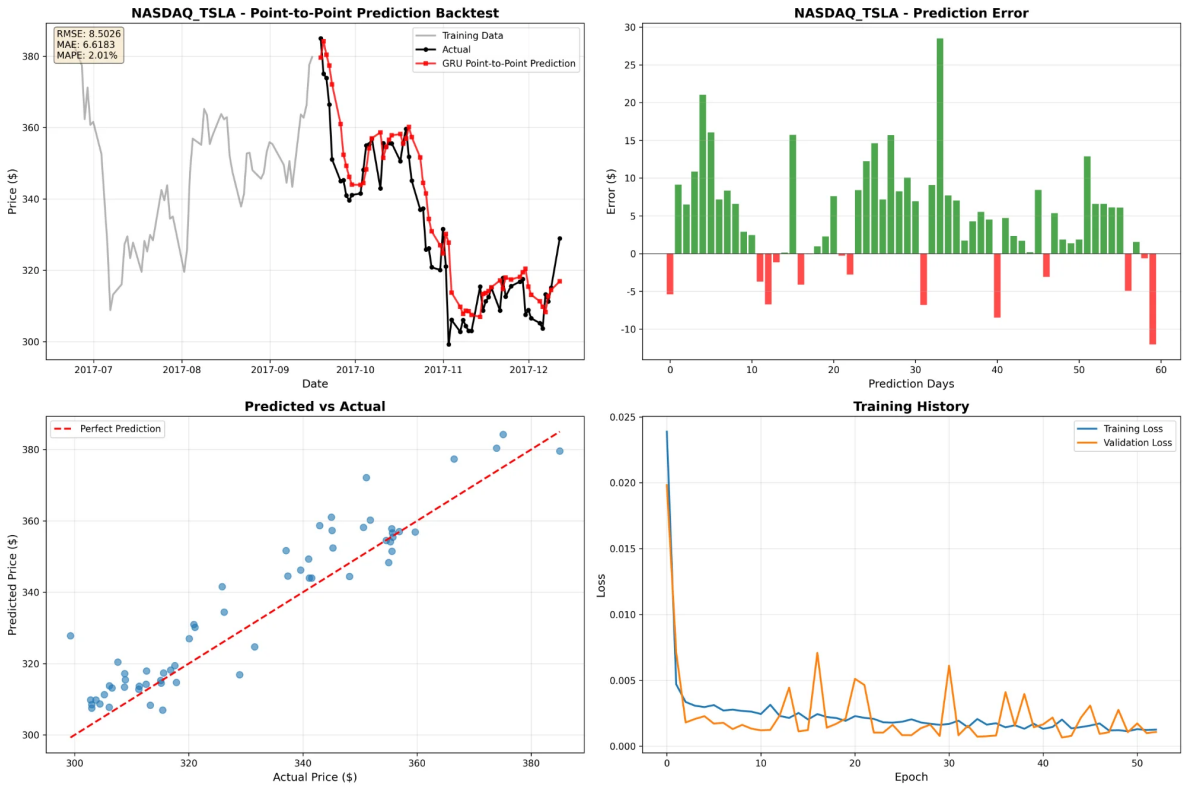

Figure 6: TSLA GRU Point-to-Point Prediction Results

6.4 Comparative Analysis

Table 1: Model Performance Ranking by MAPE

Stock	Model	MAPE
GOOGL	ARIMA	0.76%
GOOGL	GRU	1.32%
GOOGL	LSTM	1.75%
TSLA	ARIMA	1.58%
TSLA	GRU	2.01%
TSLA	LSTM	3.48%

Key findings:

1. **ARIMA Competitive for Stable Stocks**: ARIMA achieved best MAPE for GOOGL (0.76%), suggesting linear models remain effective for relatively stable, trending stocks.

2. **GRU Consistently Outperforms LSTM**: GRU demonstrated superior performance across both stocks. For GOOGL: GRU (1.32%) vs LSTM (1.75%), achieving 25% lower MAPE. For TSLA: GRU (2.01%) vs LSTM (3.48%), achieving 42% lower MAPE. The simpler gating mechanism proves advantageous for both stable and volatile stocks.

3. **Model Complexity Should Match Data Characteristics**: While ARIMA excels for stable stocks, neural networks capture local trends better during short-term fluctuations. GRU's architecture balances complexity and generalization effectively.

4. **Directional Accuracy Challenges**: All models struggled with directional prediction (48-52% accuracy), barely exceeding random chance, questioning utility of point predictions versus probabilistic forecasts.

5. **Computational Trade-offs**: ARIMA requires slow iterative refitting but no GPU; LSTM fastest inference after training; GRU balances speed and accuracy with fewer parameters.

7 Conclusion

This mid-term report documents significant progress implementing a comparative stock price prediction system. Despite pivoting from the original pre-trained model approach, the project successfully established a robust data pipeline, implemented four prediction architectures (ARIMA, LSTM, GRU, SimpleRNN), developed comprehensive evaluation and visualization tools, and created an extensible codebase with CLI interfaces.

Key empirical insights: ARIMA remains competitive for stable stocks (0.76% MAPE), GRU consistently outperforms LSTM across both stable and volatile stocks (1.32% vs 1.75% for GOOGL; 2.01% vs 3.48% for TSLA), all models struggle with directional prediction (50% accuracy), and model complexity should match data characteristics.

The project remains on schedule with core implementation complete. Completed components include data acquisition pipeline for individual stocks and S&P 500, all baseline model implementations, comprehensive evaluation metrics, visualization system, and preliminary experimental results.

8 Recommendations for Future Work

8.1 Immediate Next Steps

Extended Experimental Coverage: Expand testing to 20-30 diverse S&P 500 constituent stocks across sectors (technology, finance, healthcare, energy, consumer goods) with different market caps and volatility profiles. Evaluate performance during multiple time periods including bull markets (2019-2021), volatile periods (COVID-19 crash 2020), and bear markets (2022 correction). Conduct sensitivity analysis varying lookback windows (30, 60, 90, 120 days), train-test splits, and historical data lengths (3, 5, 10 years).

Hyperparameter Optimization: Implement systematic hyperparameter tuning for each model architecture. For ARIMA, use auto.arima equivalent with grid search over (p, d, q) parameters and AIC/BIC-based selection. For neural networks, tune layer depth (2-5

layers), units per layer (50, 100, 200), dropout rates (0.1-0.5), and learning rate scheduling using Bayesian optimization with Optuna (50-100 trials per model type).

8.2 Informer Model Implementation

Architecture Development: Implement the Informer architecture [13], a transformer-based model specifically designed for long sequence time-series forecasting. The Informer addresses computational complexity through:

1. **ProbSparse Self-Attention**: Reduces complexity from $O(L^2)$ to $O(L \log L)$ by selecting dominant queries
2. **Self-Attention Distilling**: Progressively reduces input dimensions through convolution and max-pooling
3. **Generative Style Decoder**: Predicts long sequences in one forward pass rather than step-by-step

The Informer will be trained on S&P 500 data with sequence lengths of 60-120 days, testing its ability to capture long-term dependencies that RNN-based models may miss.

Preprocessing Insights from Pre-training Literature: While full replication of the pre-training framework is infeasible due to missing implementation details, the project will selectively incorporate preprocessing techniques where documented:

- **Moving Average Features**: Calculate 5-day, 10-day, and 20-day moving averages as additional input features to capture price momentum
- **Sector Encoding**: One-hot encode stock sectors to help models learn sector-specific patterns
- **Normalization Strategies**: Compare stock-wise normalization (each stock normalized independently) versus portfolio-wise normalization (all stocks in same scale)

Important Caveat: The original pre-training paper lacks detailed specifications for data processing pipelines and training parameters. Therefore, the project focuses on traditional forecasting methods enhanced with documented preprocessing techniques, rather than attempting full transformer pre-training.

8.3 Macroeconomic Factor Integration

A key innovation of the final project will be investigating how macroeconomic indicators influence S&P 500 prediction accuracy. This addresses a practical limitation of pure price-based models that ignore fundamental economic conditions.

Target Economic Indicators:

1. **Consumer Price Index (CPI)**: Monthly inflation indicator affecting corporate earnings and Fed policy. Forward-fill to daily frequency.
2. **Long-term Interest Rate (10-Year Treasury Yield)**: Daily rates influencing discount rates for equity valuations and investor risk appetite.
3. **Price-to-Earnings (P/E) Ratio**: S&P 500 aggregate P/E ratio indicating market valuation levels and potential corrections.

Integration Methodology:

- **Feature Concatenation**: Append economic indicators as additional input features to neural networks
- **Multi-Modal Architecture**: Develop separate encoders for price sequences and economic indicators, fusing representations before prediction
- **Ablation Studies**: Systematically add/remove each indicator to quantify individual contributions to prediction accuracy

Expected Outcomes: Economic indicators are hypothesized to improve prediction accuracy during regime shifts (recessions, policy changes) when pure price patterns become unreliable. The analysis will quantify MAPE improvements and identify which indicators provide the strongest predictive signals.

8.4 Revised Scope: S&P 500 Focus

Cross-Market Analysis Abandoned: The original plan to investigate transfer learning between S&P 500 and Hang Seng Index has been revised. Instead, the project will:

1. Conduct deep analysis of S&P 500 constituents across multiple sectors
2. Compare model performance between high-cap stable stocks (e.g., AAPL, MSFT) and mid-cap volatile stocks
3. Investigate sector-specific prediction challenges (technology vs. finance vs. healthcare)
4. Analyze how economic indicators affect different sectors differently

This focused approach enables more rigorous analysis of a single market rather than superficial comparison across markets with different regulatory environments and trading hours.

8.5 Ensemble Methods

Model Combination Strategies: Combine predictions from multiple models to leverage their complementary strengths:

- **Simple Averaging**: Equal-weight or validation-performance-weighted average of ARIMA, LSTM, GRU, Informer
- **Stacking**: Train a meta-learner (gradient boosting or linear regression) on base model predictions
- **Volatility-Adaptive Weighting**: Dynamically adjust model weights based on recent volatility (favor ARIMA for low volatility, GRU for high volatility)

8.6 Final Report Deliverables

The completed project will provide:

1. **Comprehensive Model Comparison**: Statistical analysis (paired t-tests, confidence intervals) comparing ARIMA, LSTM, GRU, SimpleRNN, and Informer across 20+ S&P 500 stocks
2. **Economic Indicator Impact Analysis**: Quantified improvements from incorporating CPI, interest rates, and P/E ratios with ablation study results
3. **Sector-Specific Insights**: Analysis of prediction difficulty across technology, finance, healthcare, energy, and consumer sectors
4. **Practical Recommendations**: Guidelines for practitioners on:
   - Which model to use for different stock volatility profiles
   - When economic indicators provide meaningful improvements
   - Computational cost vs. accuracy trade-offs
   - Limitations and failure modes of each approach
5. **Open-Source Implementation**: Documented codebase with preprocessing pipelines, trained models, and evaluation scripts for reproducibility

8.7 Timeline for Remaining Work

Table 2: Remaining Project Timeline

Weeks 1-2	Extended experiments on 20+ stocks; hyperparameter optimization
Weeks 3-4	Informer implementation and training; economic data collection
Weeks 5-6	Economic indicator integration and ablation studies
Weeks 7-8	Ensemble methods; sector-specific analysis
Weeks 9-10	Final report writing; documentation; presentation preparation

9 Summary

This report documents a stock price prediction project that successfully adapted to implementation challenges with the original pre-training framework. After recognizing the absence of detailed data processing specifications and complete transformer training parameters, the project pivoted to point-to-point prediction using traditional time series models (ARIMA, LSTM, GRU, SimpleRNN).

Preliminary experiments demonstrate model performance varies significantly with stock volatility characteristics: ARIMA remains competitive for stable stocks (0.76% MAPE), while GRU consistently outperforms LSTM across both stocks (1.32% vs 1.75% for GOOGL; 2.01% vs 3.48% for TSLA). All models struggle with directional prediction (50% accuracy), highlighting fundamental challenges in market forecasting.

Revised Project Direction: The final project focuses on traditional time series models for S&P 500 prediction, enhanced with: (1) Informer transformer architecture implementation for comparison, (2) selective preprocessing techniques from recent literature, (3) integration of macroeconomic indicators (CPI, long-term interest rates, P/E ratios), and (4) comprehensive sector-specific analysis across S&P 500 constituents.

Key Clarification: The pre-training framework's code lacks complete data processing pipelines and training hyperparameters. The final project emphasizes comparative analysis of traditional forecasting methods enhanced with documented preprocessing insights, delivering practical results while maintaining academic rigor. The dataset comprises individual stock data from NASDAQ, NYSE, and NYSE American (1987-2017), plus S&P 500 data with economic indicators (2016-2026) for macroeconomic factor analysis.

The project's final phase will focus on three key innovations:

1. **Informer Implementation**: Developing a transformer-based architecture specifically designed for long sequence forecasting, incorporating documented preprocessing techniques from the pre-training literature where feasible

2. **Macroeconomic Integration**: Investigating how Consumer Price Index, Long-term

Interest Rates, and P/E Ratio influence S&P 500 prediction accuracy through feature integration and ablation studies

3. **Traditional Model Optimization**: Conducting comprehensive comparison of classical and deep learning approaches on S&P 500 constituents across multiple sectors with hyperparameter optimization and ensemble methods

The revised scope eliminates cross-market analysis (Hang Seng Index) in favor of deeper investigation of economic factors affecting a single market. This focused approach enables rigorous analysis of how macroeconomic conditions influence prediction accuracy—a practical consideration often overlooked in pure price-based models.

The methodological pivot proved beneficial, yielding a tractable problem with clear evaluation criteria and practical relevance for short-term forecasting. This work will contribute to understanding machine learning applications in quantitative finance by providing systematic empirical comparison across classical statistical and modern deep learning methods, highlighting their respective strengths, weaknesses, and appropriate application contexts—particularly when enhanced with macroeconomic indicators.

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